

Discrete Mathematics

Week 1: Propositions and Logical Operators

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Definition

A **proposition** is a declarative statement that is either **true** or **false**, but not both.

Key Points:

- Must be a declarative sentence
- Must have a definite truth value (T or F)
- Cannot be both true and false

Which of these are propositions?

Determine if each statement is a proposition:

- ① The capital city of Turkey is Istanbul.
- ② $2 + 3 = 5$
- ③ Today the weather is great.
- ④ How old are you?
- ⑤ Do not be late!!!
- ⑥ 2375
- ⑦ A compiler translates source code into machine code.
- ⑧ Is Python an object-oriented language?

Which of these are propositions? - Answers

- ① The capital city of Turkey is Istanbul. (False - Ankara is capital)
✓ Proposition
- ② $2 + 3 = 5$ (True) ✓ Proposition
- ③ Today the weather is great. × Not a proposition (subjective)
- ④ How old are you? × Not a proposition (question)
- ⑤ Do not be late!!! × Not a proposition (command)
- ⑥ 2375 ✓ Proposition (number statement)
- ⑦ A compiler translates source code into machine code. (True)
✓ Proposition
- ⑧ Is Python an object-oriented language? × Not a proposition (question)

Propositional Notation

Syntax

Propositions are represented using lowercase letters:

$$p, q, r, s, t, \dots$$

Examples:

- p : $3 + 7 = 10$
- q : This rose is red.
- r : The algorithm has $O(n^2)$ time complexity.
- s : A binary tree has at most 2^h nodes at height h .

Truth Values

Definition

Every proposition has a **truth value**:

- **True (T)** = 1
- **False (F)** = 0

Examples:

Proposition	Truth Value
$p: 5 > 3$	T (1)
$q: 2 + 2 = 5$	F (0)
$r: A \text{ stack follows LIFO principle}$	T (1)
$s: 7 \text{ is an even number}$	F (0)

Negation of a Proposition

Definition

The **negation** of proposition p is denoted by $\neg p$ or p' and has the opposite truth value of p .

Truth Table:

p	$\neg p$
1	0
0	1

Example:

- p : The capital city of Turkey is Istanbul. (F)
- $\neg p$: The capital city of Turkey is NOT Istanbul. (T)

Negation - More Examples

Examples from mathematics and computer science:

- $r: 3 = 3$ $\neg r: 3 \neq 3$
- $r = 1$ $\neg r = 0$
- $s: 3 > 5$ $\neg s: 3 \leq 5$
- $s = 0$ $\neg s = 1$
- $t: \text{The queue is empty}$ $\neg t: \text{The queue is not empty}$
- $t = 0$ $\neg t = 1$

Open Propositions (Predicates)

Definition

An **open proposition** or **predicate** is a statement containing variables. Its truth value depends on the values assigned to the variables.

Example:

$$p(x) : x^2 - 9 = 0$$

- $p(5) : 5^2 - 9 = 0 \rightarrow 25 - 9 = 0$ False (0)
- $p(3) : 3^2 - 9 = 0 \rightarrow 9 - 9 = 0$ True (1)
- $p(-3) : (-3)^2 - 9 = 0 \rightarrow 9 - 9 = 0$ True (1)

Question: Which x values make $p(x) = 1$?

$$x = 3 \text{ or } x = -3$$

Open Propositions - CS Example

Computer Science Example:

$p(x)$: “ x is a valid IPv4 address”

- $p(\text{"192.168.1.1"}) \rightarrow \text{True (1)}$
- $p(\text{"256.1.1.1"}) \rightarrow \text{False (0)}$
- $p(\text{"10.0.0.1"}) \rightarrow \text{True (1)}$

Another example:

$q(n)$: “Algorithm runs in $O(n)$ time”

- $q(\text{linear search}) \rightarrow \text{True (1)}$
- $q(\text{bubble sort}) \rightarrow \text{False (0)}$

Logical Operators (Connectives)

Definition

When we connect propositions using **logical operators**, we create **compound propositions**.

Five Main Logical Operators:

- 1 $\wedge$ - AND (Conjunction)
- 2 $\vee$ - OR (Disjunction)
- 3 $\rightarrow$ - If...then... (Implication)
- 4 $\leftrightarrow$ - If and only if (Biconditional)
- 5 $\oplus$ - Exclusive OR (XOR)

AND Operator ($\wedge$) - Conjunction

Definition

$p \wedge q$ is true only when **both** p and q are true.

Truth Table:

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

Example: “Today I will go to the cinema **and** cafe.”

- p : I will go to the cinema
- q : I will go to the cafe
- $p \wedge q$: Both must be true for the statement to be true

Properties of AND ($\wedge$)

Key Properties:

- 1 **Commutative:** $p \wedge q = q \wedge p$
- 2 **Idempotent:** $p \wedge p = p$
- 3 **Identity:** $p \wedge 1 = p$
- 4 **Domination:** $p \wedge 0 = 0$ (0 is dominant)

Computer Science Example:

- p : User is authenticated
- q : User has admin privileges
- $p \wedge q$: User can access admin panel

Both conditions must be satisfied!

OR Operator ($\vee$) - Disjunction

Definition

$p \vee q$ is true when **at least one** of p or q is true.

Truth Table:

p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0

Example: “Today I will go to the cinema **or** cafe.”

- True if I go to cinema, or cafe, or both

Properties of OR ($\vee$)

Key Properties:

- ① **Commutative:** $p \vee q = q \vee p$
- ② **Idempotent:** $p \vee p = p$
- ③ **Identity:** $p \vee 0 = p$
- ④ **Domination:** $p \vee 1 = 1$ (1 is dominant)
- ⑤ **Negation:** $p \vee \neg p = 1$ (Always true)

Computer Science Example:

- p : File exists in cache
- q : File exists in database
- $p \vee q$: File is accessible

File can be retrieved if it's in **either** location!

Common Properties of AND & OR

Distribution Laws:

$$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$$

De Morgan's Laws:

$$\neg(p \wedge q) = \neg p \vee \neg q$$

$$\neg(p \vee q) = \neg p \wedge \neg q$$

These laws are fundamental for simplifying logical expressions!

Implication ($\rightarrow$)

Definition

$p \rightarrow q$ means "If p then q "

- p : hypothesis (antecedent)
- q : conclusion (consequent)

Truth Table:

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

Note: Only false when p is true but q is false!

Implication - Example

Example:

$p \rightarrow q$: “If today is Saturday, then tomorrow is Sunday”

Analysis:

- When is this statement **false**?
 - Only when today IS Saturday but tomorrow is NOT Sunday
- When is this statement **true**?
 - Today is Saturday AND tomorrow is Sunday
 - Today is NOT Saturday (regardless of tomorrow)

Key insight: If the hypothesis is false, the implication is automatically true!

Properties of Implication ($\rightarrow$)

Key Properties:

- ① $p \rightarrow p = 1$ (Always true)
- ② $p \rightarrow 1 = 1$ (Always true)
- ③ $p \rightarrow 0 = \neg p$
- ④ $0 \rightarrow p = 1$ (Always true)
- ⑤ $1 \rightarrow p = p$

Computer Science Example:

- p : Input is valid
- q : Program executes without errors
- $p \rightarrow q$: “If input is valid, then program executes correctly”

Related Conditionals

Given $p \rightarrow q$, we can form three related statements:

- ① **Original:** $p \rightarrow q$
- ② **Converse:** $q \rightarrow p$ (swap p and q)
- ③ **Inverse:** $\neg p \rightarrow \neg q$ (negate both)
- ④ **Contrapositive:** $\neg q \rightarrow \neg p$ (swap and negate)

Important: The original and contrapositive are **logically equivalent!**

Proving Logical Equivalence

Theorem: $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

Proof using Truth Tables:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
1	1	0	0	1	1
1	0	0	1	0	0
0	1	1	0	1	1
0	0	1	1	1	1

Conclusion: Since the columns for $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are identical, they are **logically equivalent!**

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Related Conditionals - Example

Original: “If today is Saturday, then tomorrow is Sunday”

Converse: “If tomorrow is Sunday, then today is Saturday”

- Not necessarily true! (could be Friday)

Inverse: “If today is NOT Saturday, then tomorrow is NOT Sunday”

- Not necessarily true!

Contrapositive: “If tomorrow is NOT Sunday, then today is NOT Saturday”

- This IS true! (logically equivalent to original)

Understanding Implication: The Tricky Part

Question: Why is $p \rightarrow q$ true when p is false?

Real-life Example:

"If I take the bus to Adana, then I will arrive in Adana."

Let's analyze all cases:

- **Case 1:** I took the Adana bus $\wedge$ I arrived in Adana $\rightarrow$ **True**
- **Case 2:** I took the Adana bus $\wedge$ I arrived in Ankara $\rightarrow$ **False**
- **Case 3:** I took the Ankara bus $\wedge$ I arrived in Adana $\rightarrow$ **True**
- **Case 4:** I took the Ankara bus $\wedge$ I arrived in Ankara $\rightarrow$ **True**

Key Insight: The promise is only broken in Case 2! If I don't take the Adana bus (hypothesis is false), the statement makes no claim about where I'll end up, so it cannot be false.

Implication in Computer Science

Promise: “If you enter valid credentials, then you will be logged in.”

$$p \rightarrow q$$

- p : You enter valid credentials
- q : You are logged in

When is this promise broken?

Credentials	Result	Promise Status
Valid	Logged in	Kept
Valid	NOT logged in	BROKEN
Invalid	Logged in	Not broken (no promise made)
Invalid	NOT logged in	Not broken (no promise made)

Conclusion: The implication only makes a promise when the hypothesis is true!